

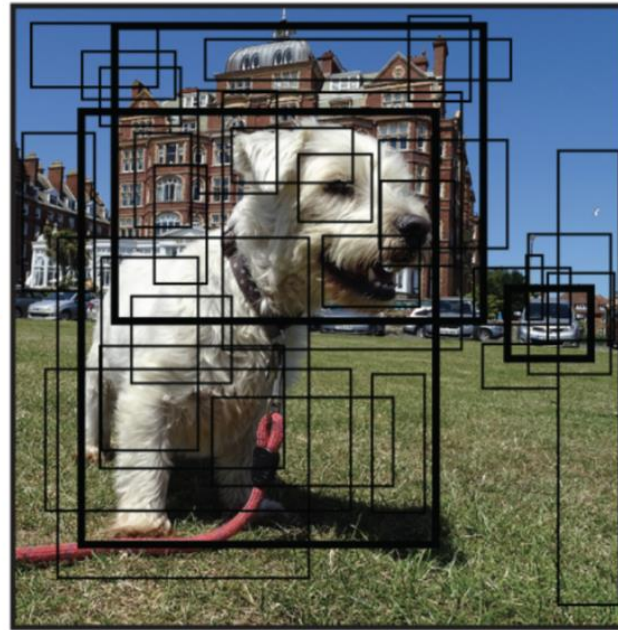
Lecture 18

Evolution of YOLO Models

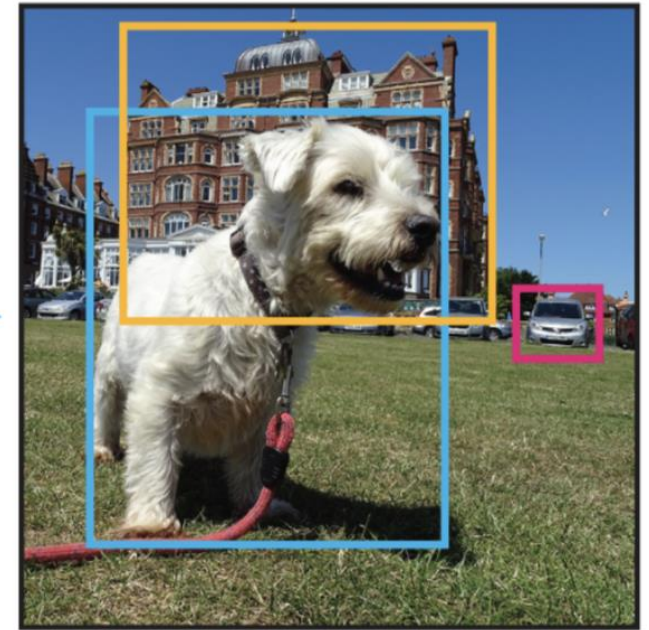
You Look Only Once (YOLO)



Splits the image into grids



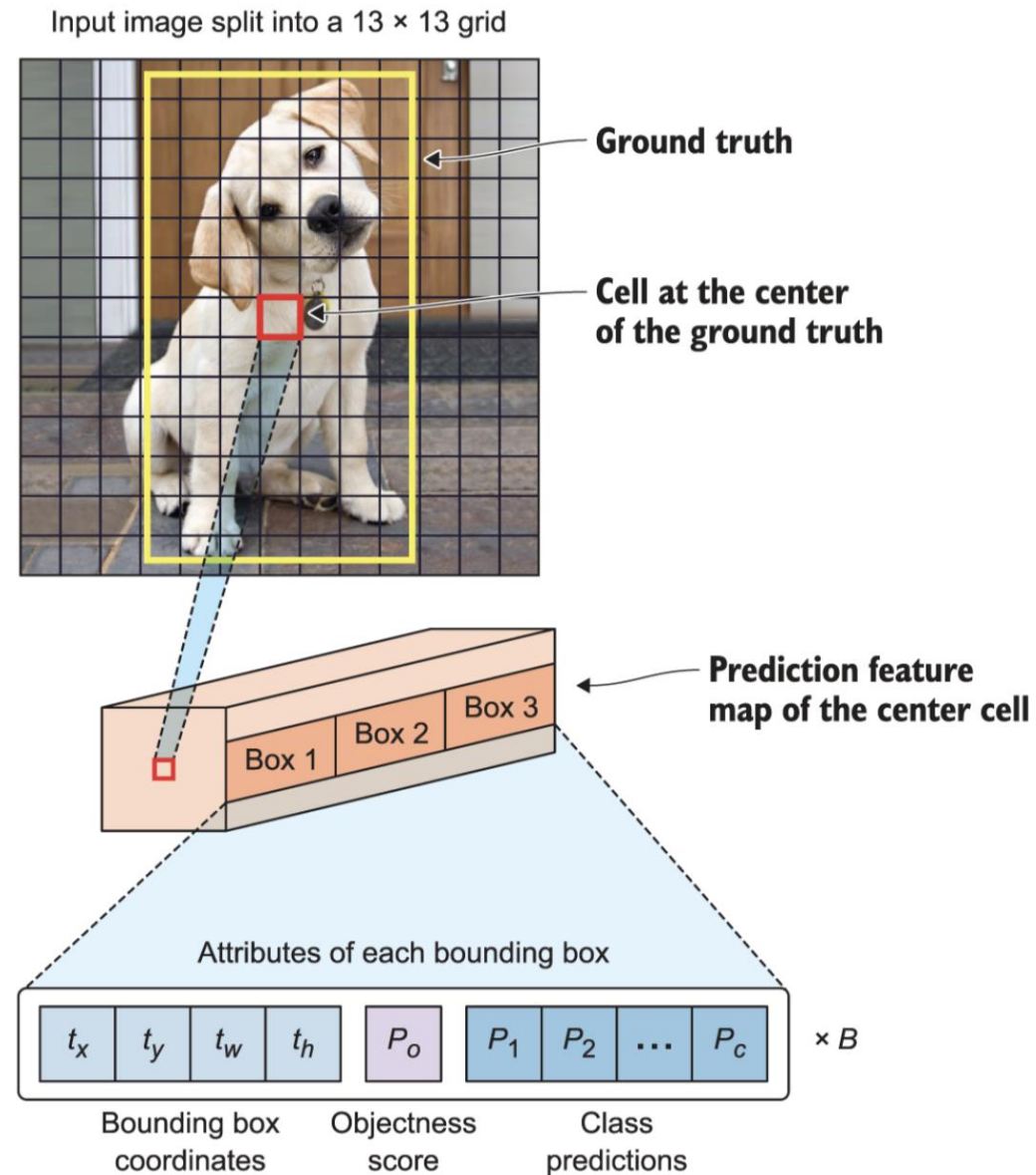
Predicts bounding boxes
and classifications



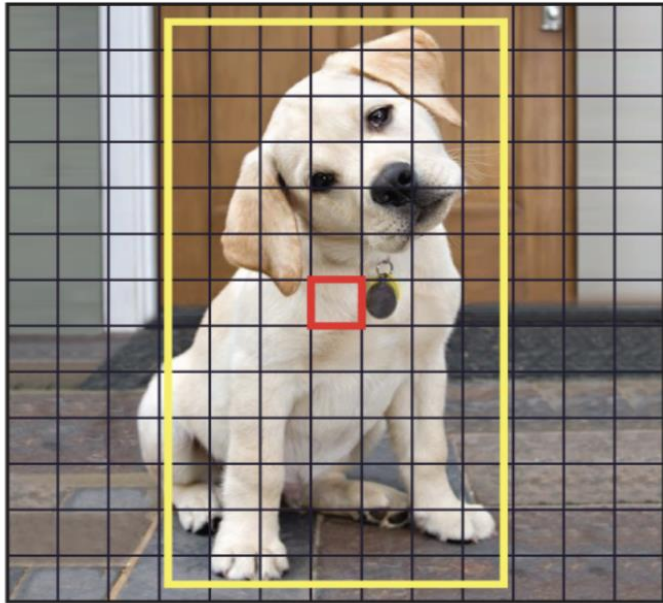
Final predictions after
non-maximum suppression

YOLO workflow

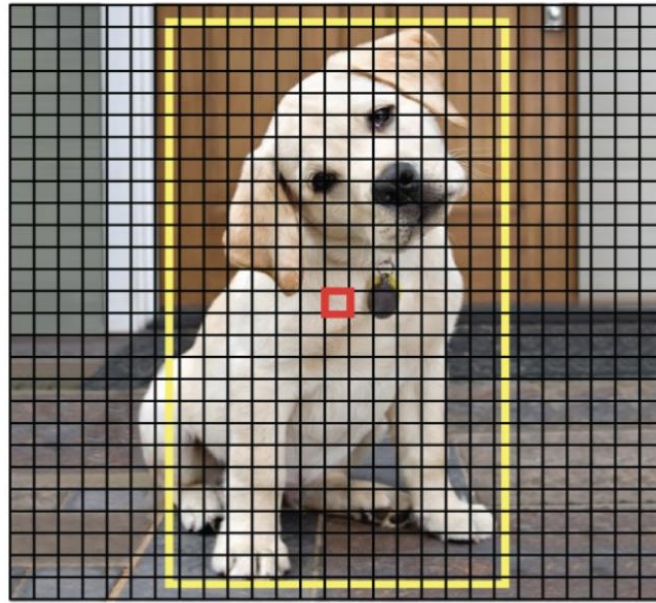
- Example of a YOLOv3 workflow when applying a 13×13 grid to the input image.
- The input image is split into 169 cells.
- Each cell predicts B number of bounding boxes and their objectness score along with their class predictions.
- In this example, we show the cell at the center of the ground-truth making predictions for 3 boxes ($B = 3$).
- Each prediction has the following attributes: bounding box coordinates, objectness score, and class predictions



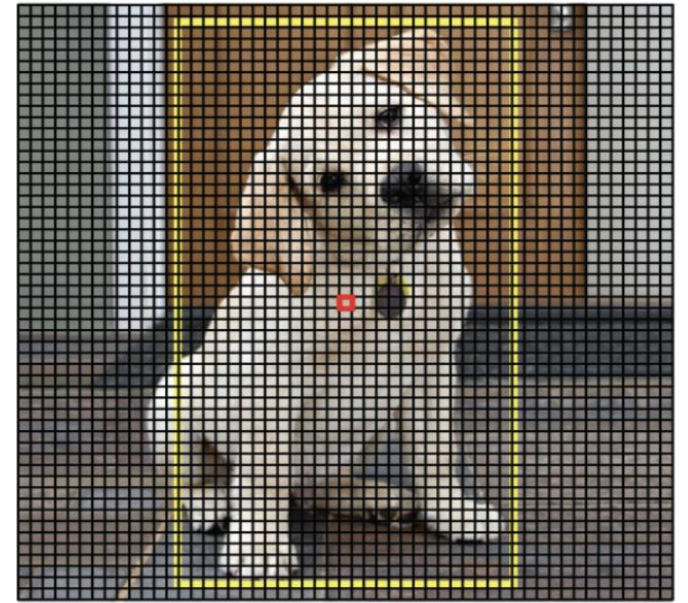
Prediction feature maps at varied scales



13×13

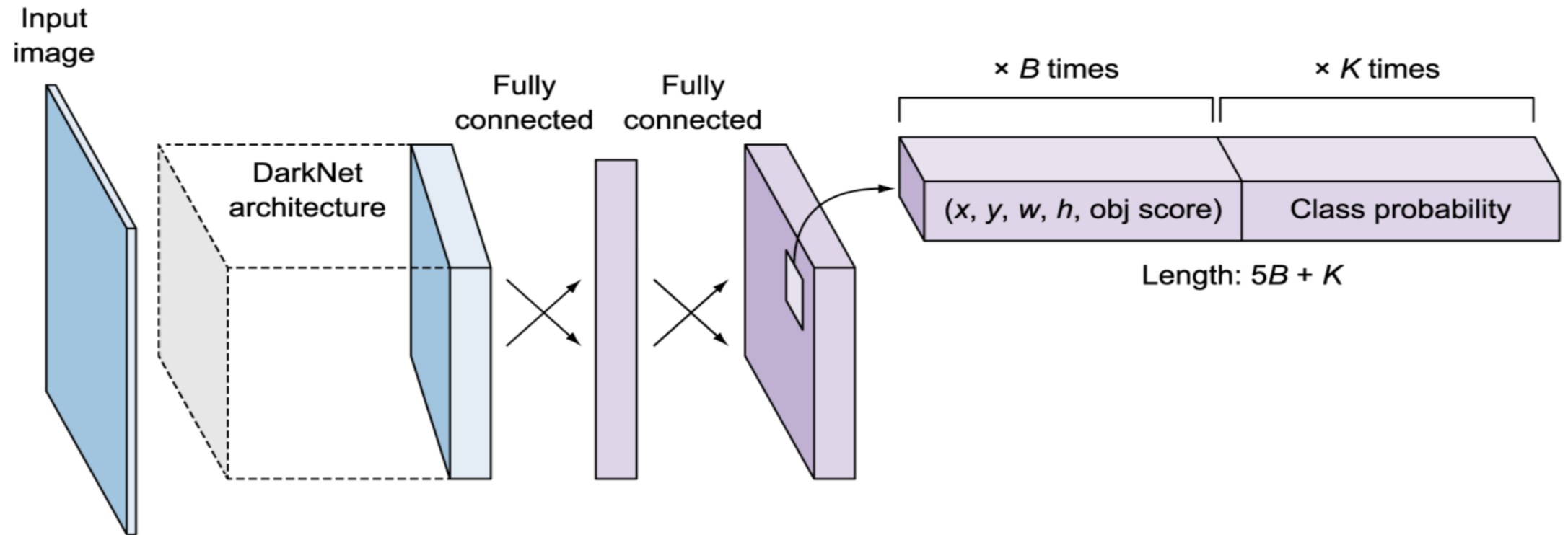


26×26

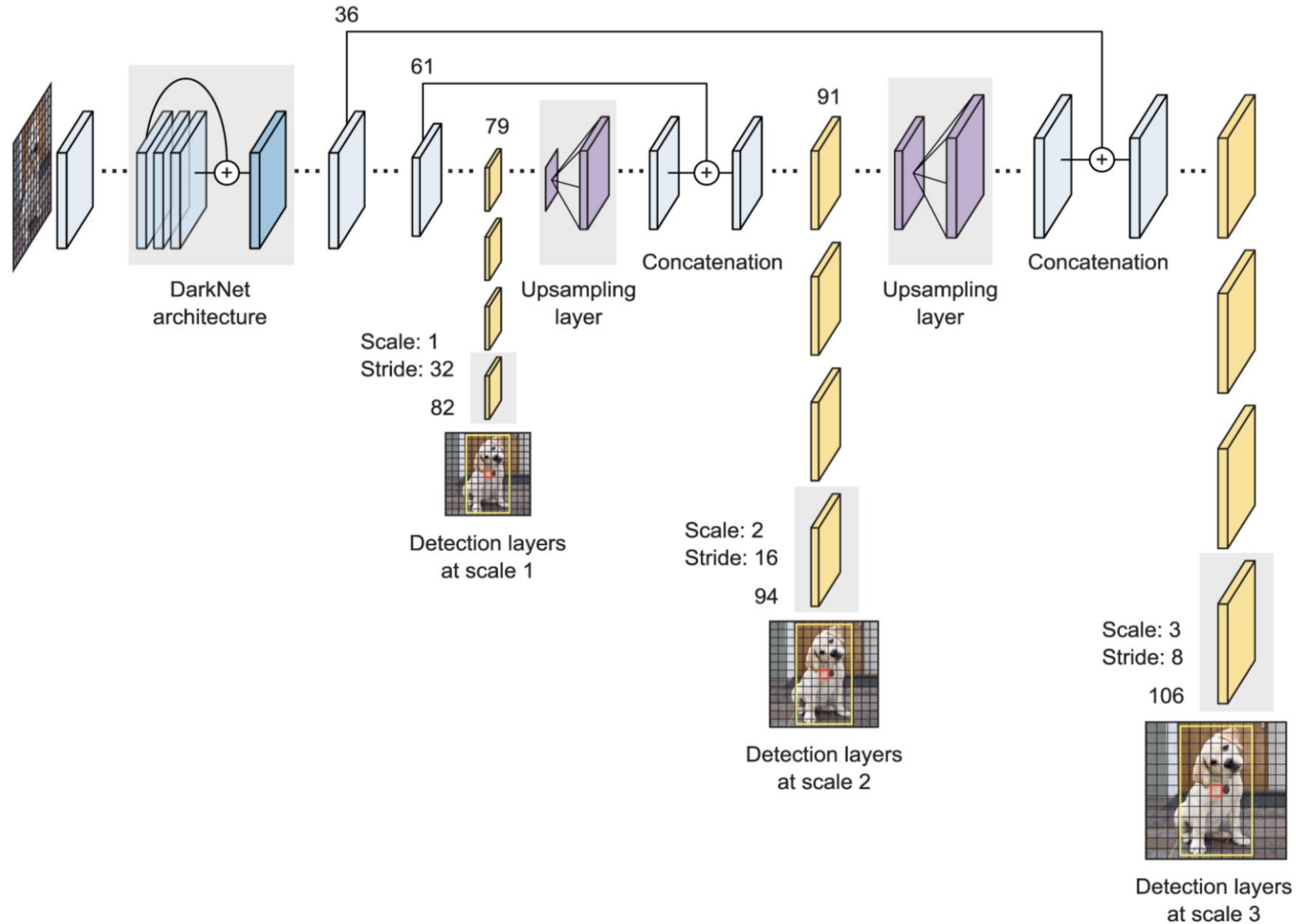


52×52

YOLO Architecture



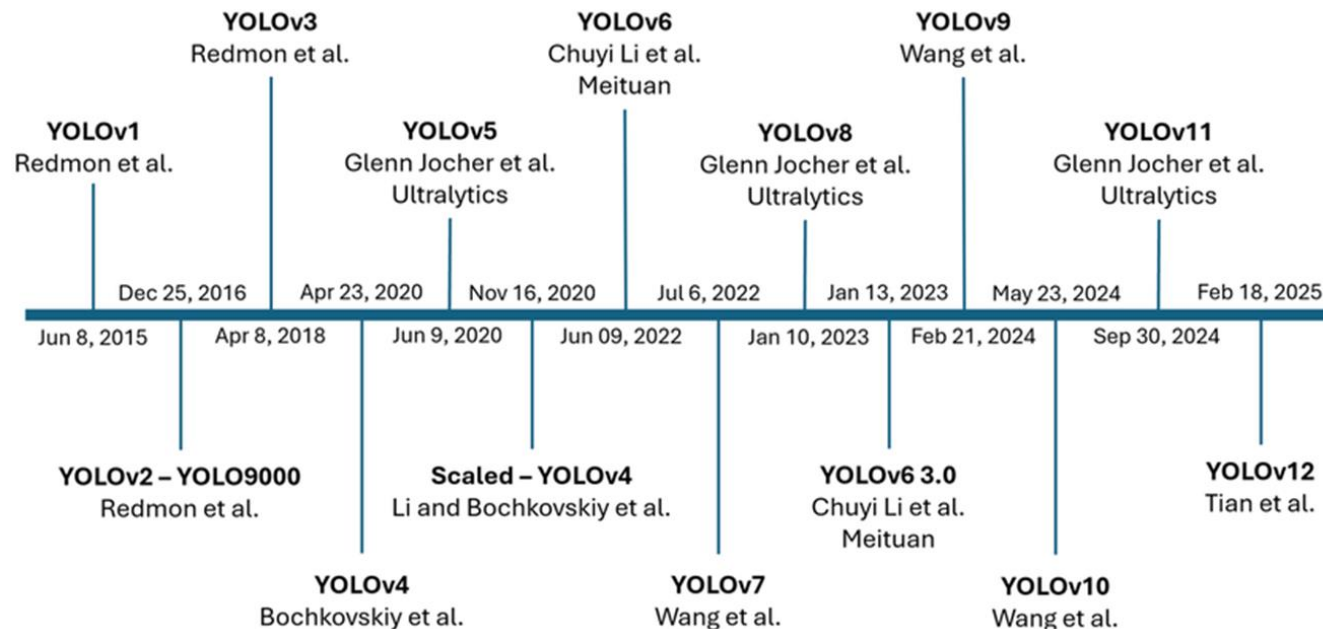
YOLO Network Architecture



YOLO model versions

You only look once -> Single stage object detection model

- YOLO predicts boxes + classes in one forward pass, enabling real-time inspection speeds.
- Good accuracy–efficiency tradeoff: YOLO offers lightweight variants suitable for factory/edge deployment while keeping strong accuracy.
- A family of versions (v1 → v12+): newer releases refine feature extraction, optimization, and efficiency



YOLO versions

Version	Year	Key Developer	Main Innovation
YOLOv1	2015	Joseph Redmon	Single-shot detection grid
YOLOv2	2016	Joseph Redmon	Anchor boxes, batch norm
YOLOv3	2018	Joseph Redmon	Multi-scale, Darknet-53 kili-technology
YOLOv4	2020	Alexey Bochkovskiy	CSPDarknet, better accuracy
YOLOv5	2020	Ultralytics	PyTorch, easy training
YOLOv8	2023	Ultralytics	Segmentation, pose support
YOLOv10	2024	Tsinghua Univ.	NMS-free, edge-optimized
YOLOv11	2024	Ultralytics	Hybrid CNN-transformer
YOLO26	2026	Ultralytics	Edge deployment, NMS-free inference

Performance Comparison with Metrics

Aspect	Early (v1-v4)	Modern (v8+)	YOLO26
Speed	Moderate	High	Edge-optimized
Accuracy (mAP)	60-70%	80-90%	SOTA on COCO
Size	Large	Compact	Smaller models
Tasks	Detection	+Seg/Pose	All + OBB

Yolov26: The Edge-Optimized Powerhouse

- End-to-End, NMS-Free Real-Time Object Detection
- Released in Early 2026
- Source: Ultralytics (Glenn Jocher & Jing Qiu)

Yolov26

- **The Next Gen Vision AI:** Released Jan 2026, this is a generational leap in deployment efficiency.
- **Core Goal:** "Edge-First" redesign designed for real-time deployment on IoT, drones, and CPU-only devices.
- **Key Advantage:** Achieves up to 43% faster CPU inference compared to previous versions.

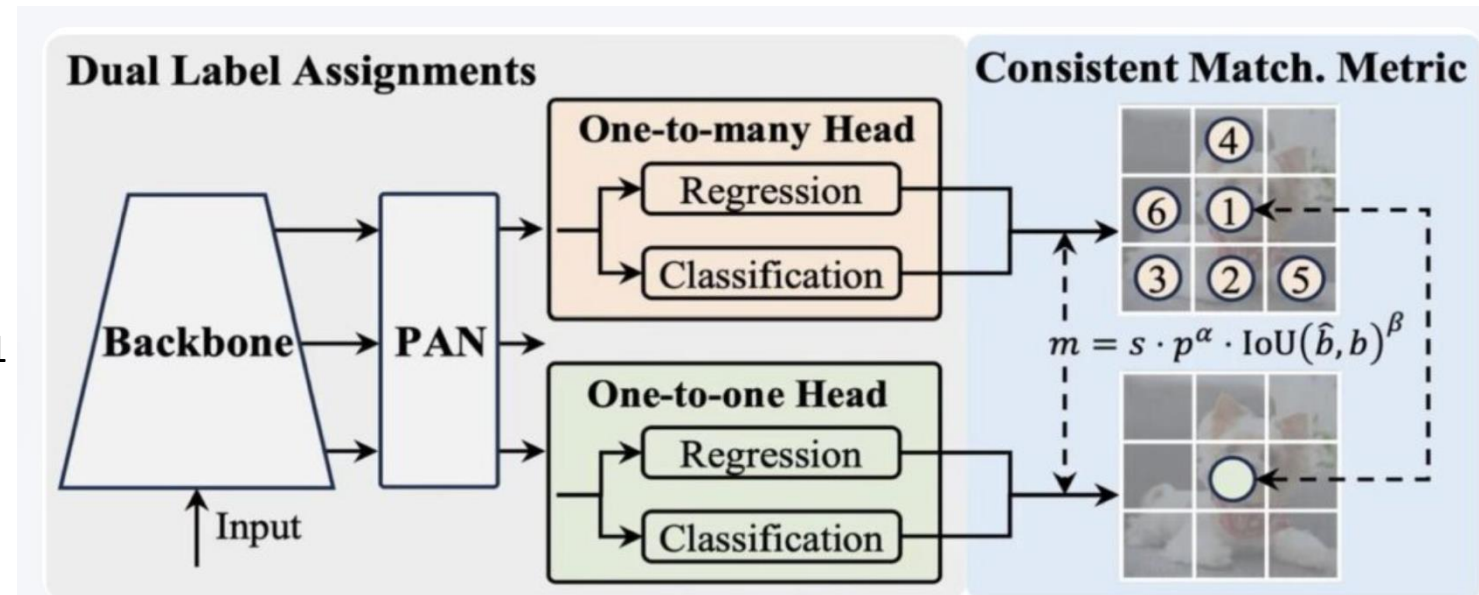
Key Architectural Advancements

- **NMS-Free Design:** Eliminates Non-Maximum Suppression (NMS), allowing for faster, simpler, and deterministic end-to-end inference.
- **DFL Removal:** Removed Distribution Focal Loss (DFL) to increase speed and improve compatibility with edge hardware (quantization stability).
- **MuSGD Optimizer:** A new hybrid optimizer (SGD + Muon) for faster convergence and more stable training.

NMS-Free Design

- **Dual-Head Training Mechanism (1-to-1 Matching)**

- **The "Dual Assignment" Approach:** During training, YOLO26 uses two different "heads" to supervise the backbone network.
- **One-to-Many Head:** Acts like traditional YOLO, assigning multiple positive anchor points to a single ground-truth object, providing rich gradients for learning.
- **One-to-One Head (NMS-Free):** Uses a strict top-1 matching strategy during training. It forces the model to select only the single best predicted box for each object, teaching the model to be decisive.
- **Consistency Metric:** A "Consistent Matching Metric" forces the One-to-One head to align with the best predictions from the One-to-Many head. As a result, at inference time, the model naturally outputs one box per object without needing to filter duplicates.



DFL Removal

- DFL - a technique used in object detection (notably YOLOv8) to improve bounding box localization by predicting a probability distribution of box edges, rather than a single direct coordinate.
 - It optimizes the network to focus on values near the target, enhancing accuracy for blurry or ambiguous object boundaries.
 - Unlike traditional regression that minimizes the distance between predicted and ground-truth coordinates, DFL works by taking a set of predicted probabilities for adjacent pixels and minimizing the loss of the two values closest to the target.
 - It is particularly effective for small objects or partial occlusions, where precise localization is difficult.
- Previous models (YOLOv8-YOLO11) used Distribution Focal Loss (DFL) to model object boundaries as probability distributions. While precise, DFL requires expensive, non-linear softmax operations that are hard to run on edge CPUs.
- **Direct Prediction:** YOLO26 removes DFL, simplifying the detection head to output final box coordinates (x, y, w, h) directly. This "direct regression" removes complex decoding math, accelerating inference on embedded devices.

MuSGD Optimizer

- Inspired by Large Language Model (LLM) training (specifically Moonshot AI's Kimi K2), YOLO26 uses a hybrid optimizer called MuSGD (Momentum-Unified SGD). This combines standard SGD with matrix orthogonalization (Muon), ensuring stable, faster convergence.
- **ProgLoss:** Progressive Loss Balancing dynamically adjusts the weights between classification and regression losses during training. It ensures that the model focuses on learning high-level features first and then refines the precise bounding box locations, maintaining precision without DFL.

Training & Detection Enhancements

- **ProgLoss (Progressive Loss Balancing):** Dynamically adjusts training to improve convergence.
- **STAL (Small-Target-Aware Label Assignment):** Drastically improves detection of small, fast-moving objects (vital for drones).
- **Specialized Angle Loss:** Improves Oriented Bounding Box (OBB) accuracy by fixing boundary issues.

Performance Benchmarks (2026)

- **Superior Efficiency:** YOLO26n (Nano) achieves ~40.9 mAP compared to YOLOv5n's ~28.0, while being significantly faster.
- **Balanced Profile:** Superior accuracy-latency profile compared to YOLOv10 and RT-DETR.
- **Task Support:** Native support for Detection, Segmentation, Pose, OBB, and Classification.

When to Choose YOLO26

- **Edge/IoT Deployment:** Best choice for Raspberry Pi, Coral NPUs, and mobile devices.
- **Drone/Aerial Analysis:** Enhanced tiny object detection via STAL.
- **Latency-Critical Apps:** Robotics and Manufacturing QC, where NMS-free allows predictable response times.

Reference

- Book – Deep Learning for Vision Systems by Mohamad Elgendy, Manning Publication, November 2020
- YOLOv26, <https://docs.ultralytics.com/models/yolo26/>